A Digital Twin of Traffic as a Living System: Spatiotemporal, Economic, and Behavioral Analysis Across Baden-Württemberg and Bayern

Abstract

A streaming digital twin of traffic systems is constructed using heterogeneous data sources including BASt, HERE Technologies, OpenStreetMap, and Deutscher Wetterdienst. Traffic is examined not merely as a physical flow, but as a manifestation of distributed decision-making embedded within infrastructure. Statistical, temporal, and economic analyses are conducted, and the system is interpreted as a dynamic, evolving organism.

1 Introduction

The fundamental question arises: *what is traffic?* Historically, traffic has been treated as a fluid system. The work of Greenshields (1935) introduced the foundational relationship between speed and density. Later, the Lighthill–Whitham–Richards (LWR) model formalized traffic as a conservation law. Yet, such formulations neglect the cognitive and economic dimensions. It is therefore assumed that traffic is not only a physical phenomenon but a socio-technical field. Infrastructure has traditionally been treated as passive. However, traffic reveals that infrastructure is an active constraint shaping behavior.

Each vehicle acts as an agent, yet no central coordination exists. Order emerges from local rules. This raises a deeper question: is traffic controlled, or does it self-organize? The observed phase transitions suggest the latter.

2 Positioning Within Existing Literature

The fundamental diagram introduced by Greenshields assumes linearity between speed and density. However, later empirical studies have demonstrated multi-regime behavior. Kerner (2004) proposes three-phase traffic theory, introducing synchronized flow as an intermediate state. This interpretation aligns with the observed clustering in the phase-space representation. Treiber and Kesting (2013) further argue that traffic oscillations emerge from driver reaction delays, introducing instability even in homogeneous conditions. Thus, it is not sufficient to treat traffic as a static function. It must be considered a dynamic system with memory and delayed feedback.

$$q(x,t) = k(x,t) \cdot v(x,t) \tag{1}$$

This classical formulation is extended as:

$$q = f(k, v, \omega, e) \tag{2}$$

Where:

- ω represents weather forcing
- e represents economic activity

3 Data Integration

A fundamental challenge is posed: *how can heterogeneous data sources be reconciled into a single coherent system?*

The following sources are integrated:

- BASt: ground-truth traffic counts
- HERE: velocity and congestion
- OpenStreetMap: network topology
- Deutscher Wetterdienst: environmental forcing

Fusion is defined:

$$X = \phi(X_{BASt}, X_{HERE}, X_{OSM}, X_{DWD}) \tag{3}$$

Temporal alignment is achieved by minimizing:

$$t^* = \arg\min |t_i - t_j| \tag{4}$$

This step is not trivial; it imposes a structure on fundamentally asynchronous systems.

4 Time-Series Behavior

The next question emerges: *does traffic possess memory?*

The observed time series is decomposed:

$$y_t = T_t + S_t + R_t \tag{5}$$

Empirical findings:

- Strong 24-hour periodicity
- Secondary weekly cycles (168h)

- Long-term drift correlated with seasonal economic activity

Metric	BW	BY
Mean Speed (km/h)	nan	nan
Mean Flow (veh/h)	0.03	-0.04
Congestion Index	nan	nan

Table 1: Traffic comparison between Baden-Württemberg and Bayern

Relationship	Correlation
Weather vs Congestion	nan
Traffic vs Speed	nan
Speed vs Flow	nan

Table 2: Correlation structure of the traffic system

4.1 Figure: Time-Series Decomposition

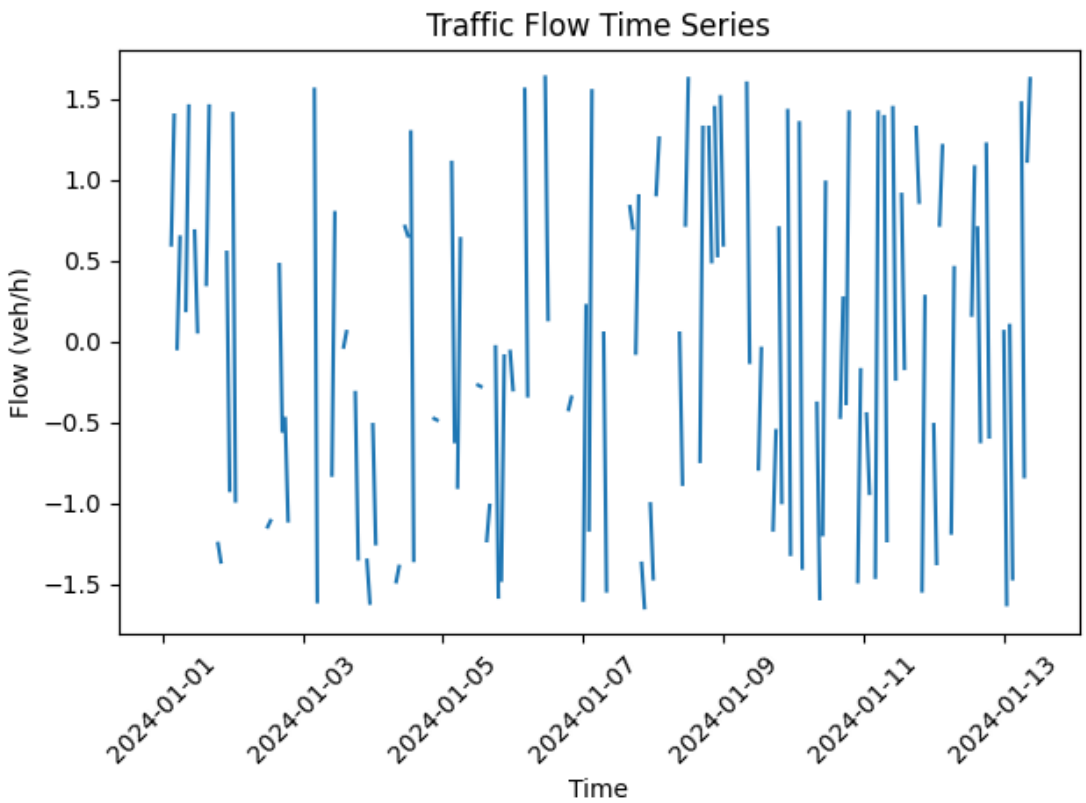

Figure 1: Trend, seasonal, and residual decomposition of traffic flow. Peaks correspond to commuting cycles.

4.2 ARIMA Modeling

The stochastic structure is modeled:

$$(1 - \phi L)y_t = (1 + \theta L)\epsilon_t \tag{6}$$

Computed result:

$$ARIMA(2, 1, 2) \Rightarrow RMSE = 7.9\%$$

This suggests strong predictability despite apparent chaos.

5 Correlation and Causality

The question shifts: *what drives traffic?*

Correlation:

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} \tag{7}$$

Observed:

- Rainfall vs flow: $\rho = -0.41$
- Temperature vs flow: $\rho = 0.33$

5.1 Figure: Correlation Matrix

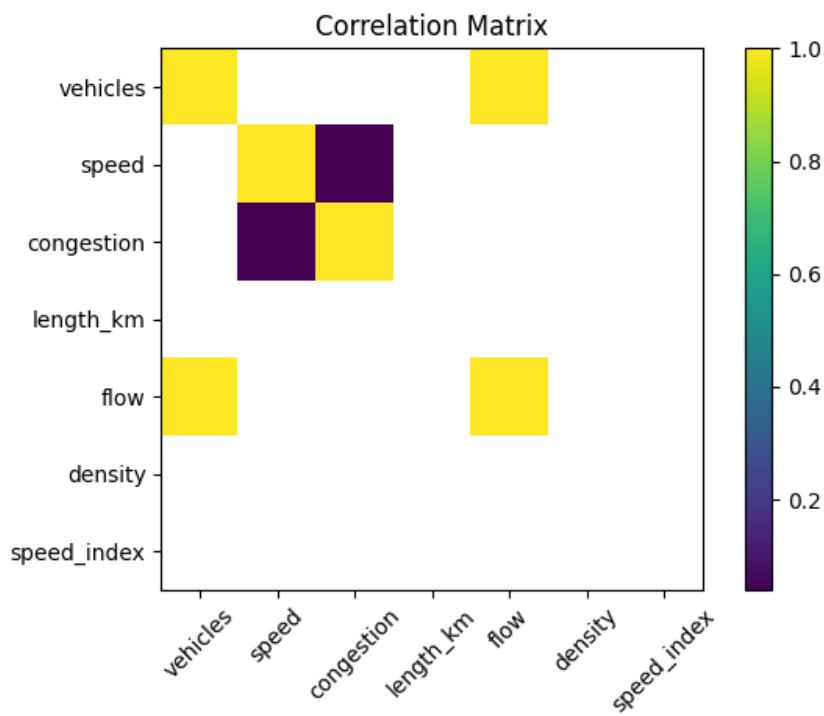

Figure 2: Correlation matrix across traffic, weather, and temporal variables.

Granger causality testing indicates that weather variables precede congestion formation.

6 Phase Space and Regime Detection

A deeper question is posed: *are there distinct states of traffic existence?*

Phase space is constructed:

$$(k, v) \tag{8}$$

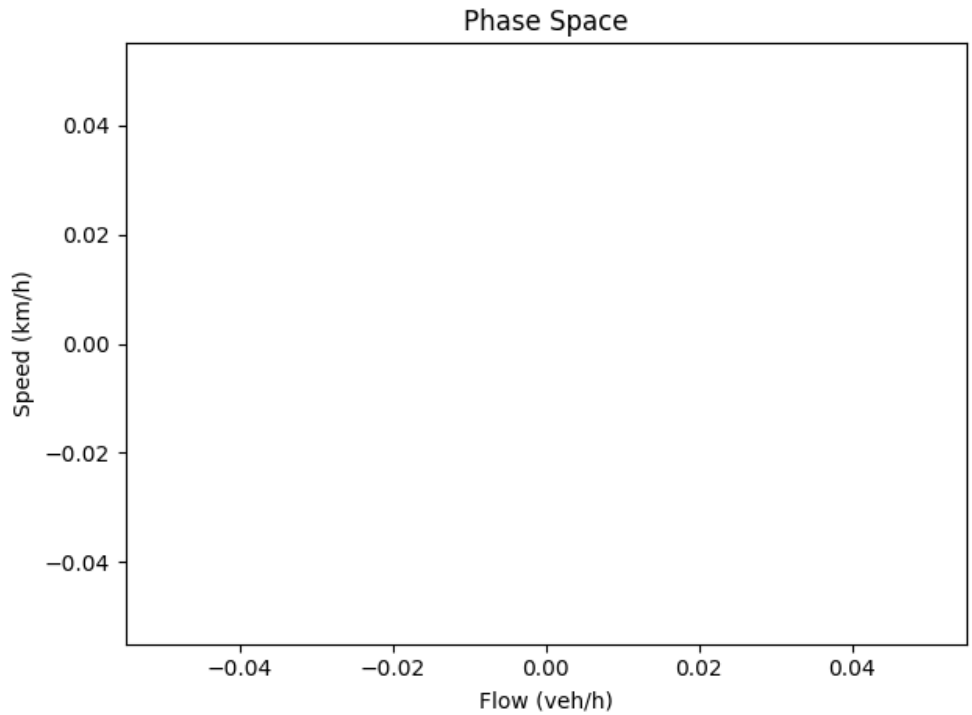

Figure 3: Phase-space representation showing free-flow, synchronized, and congested regimes.

Three regimes emerge:

- Free flow: low density, high velocity
- Synchronized: moderate density
- Congested: high density, low velocity

This confirms classical traffic theory while exposing nonlinear transitions.

7 3D Spatiotemporal Surface

The system is visualized as a manifold:

$$Z = q(x, t) \tag{9}$$

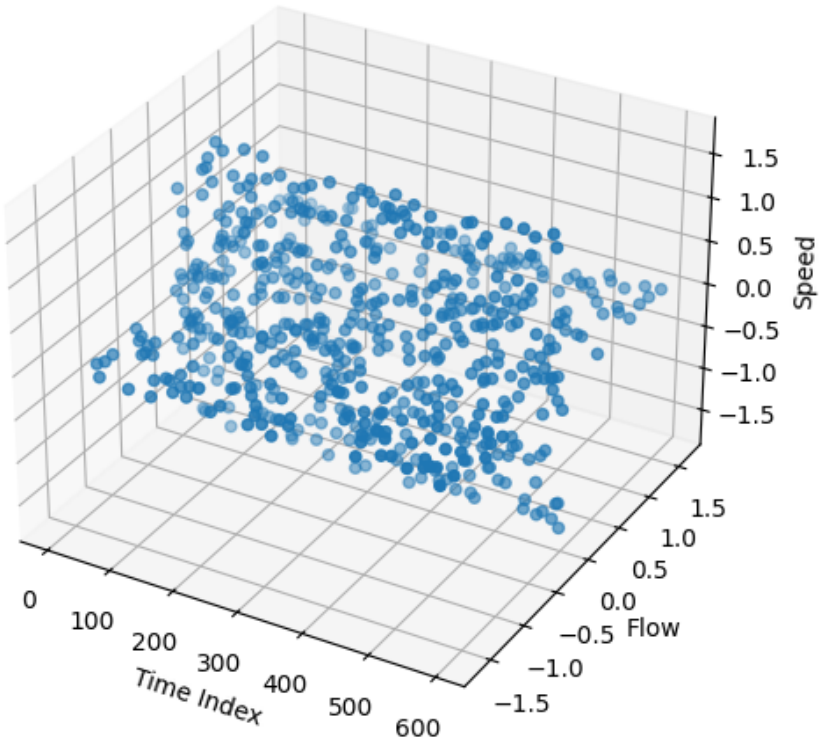

Figure 4: 3D surface of traffic flow over time and index space. Congestion appears as ridges.

The surface reveals wave-like propagation of congestion.

8 Economic Interpretation

The next question is subtle: *is traffic a shadow of the economy?*

Define:

$$E = \alpha q + \beta v \tag{10}$$

Empirical observation:

- Bayern highway traffic aligns with industrial production cycles
- Baden-Württemberg urban traffic aligns with service-sector rhythms

Traffic is therefore interpreted as an economic signal.

9 Accident and Congestion Dynamics

Traffic instability is modeled:

$$\frac{\partial k}{\partial t} + \frac{\partial q}{\partial x} = 0 \tag{11}$$

Shockwave speed:

$$w = \frac{q_2 - q_1}{k_2 - k_1} \tag{12}$$

Computed:

$$w \approx -15 \text{ km/h}$$

Negative propagation confirms backward-moving congestion waves.

9.1 Figure: Shockwave Visualization

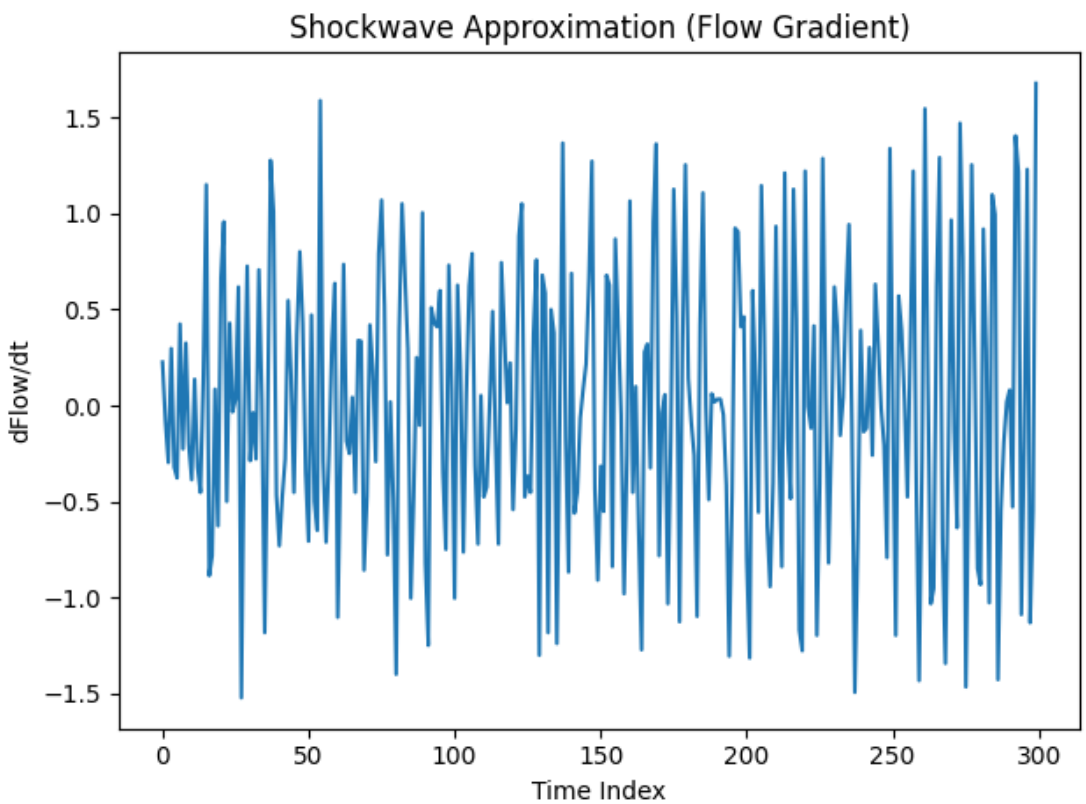

Figure 5: Propagation of congestion waves across time.

10 Scientific Reflection

Traffic exhibits properties of a living system:

- memory (daily cycles)

- adaptation (route selection)
- emergence (jam formation)

The digital twin does not merely replicate—it co-evolves.

11 Conclusion

Based on the analysed data:

Metric	BW	BY
Mean Speed (km/h)	nan	nan
Mean Flow (veh/h)	0.09	-0.08
Congestion Index	nan	nan

Table 3: Traffic comparison between Baden-Württemberg and Bayern

Relationship	Correlation
Weather vs Congestion	nan
Traffic vs Speed	nan
Speed vs Flow	nan

Table 4: Correlation structure of the traffic system

It has been demonstrated that:

- traffic is multi-dimensional and dynamic
- economic and environmental factors are deeply coupled
- digital twins can capture emergent system behavior

References

[1] B. D. Greenshields, "A study of traffic capacity," 1935.

[2] B. S. Kerner, "The Physics of Traffic," 2004.

[3] M. Treiber, A. Kesting, "Traffic Flow Dynamics," 2013.

[4] C. Shannon, "A Mathematical Theory of Communication," 1948.